

Multi-state models 1

Hein Putter

Department of Biomedical Data Sciences
Leiden University Medical Center

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Outline

Introduction

- Why multi-state models?
- Examples

Terminology, data, and likelihood

- Terminology
- Likelihood
- Data representation

Non-parametric estimation

- Theory
- Examples

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Why multi-state models? I

Multiple endpoints

- ▶ Competing risks model to describe different failures

Intermediate events

- ▶ Intermediate events may influence outcome
 - ▶ Clinical events: disease recurrence, recovery, adverse events, response, disability
 - ▶ Treatments
- ▶ Studying these intermediate events may give important information
 - ▶ Effect of treatment/covariates on intermediate events and on endpoint before or after intermediate event
 - ▶ Effect of the occurrence of intermediate events themselves

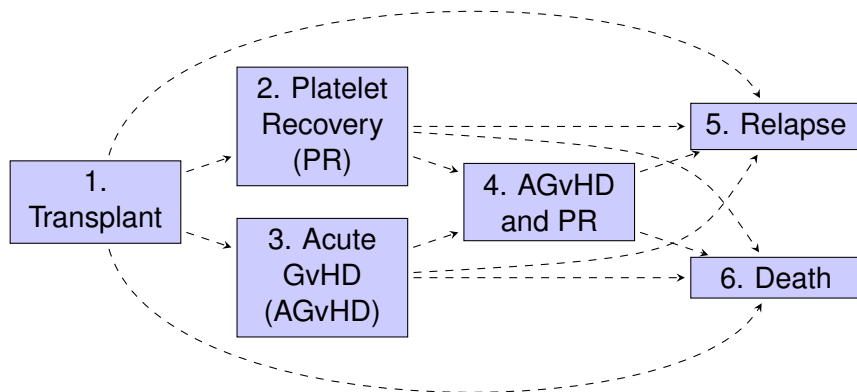
Why multi-state models? II

General objectives

- ▶ To model the disease/recovery process after stem cell transplantation, and how covariates effect this process
 - ▶ Estimation $\Rightarrow$ biology
- ▶ To give a prognosis for a patient with a given post-transplant event-history
 - ▶ Prediction $\Rightarrow$ clinical practice

Same events can serve as endpoints and intermediate events

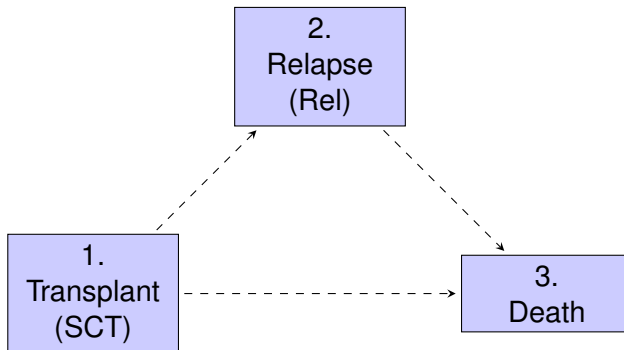
An example multi-state model



Leading example

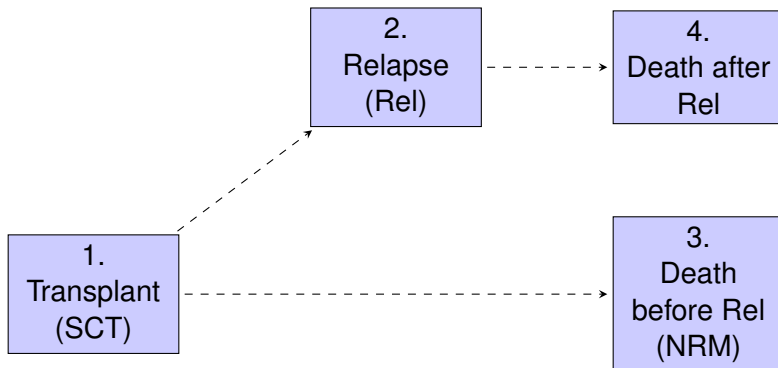
- ▶ Outcome after allogeneic stem cell transplantation of chronic lymphocytic leukemia (CLL) patients
- ▶ Data collected by European Society for Blood and Marrow Transplantation (EBMT); CLL Data Quality Initiative
- ▶ Failures of interest: death, relapse
- ▶ Covariates of interest: donor category, remission status, age, year of SCT, combination of sex recipient-donor

Illness-death model



- Note: without the Rel → Death transition, this is a competing risks model

Extended illness-death model



- ▶ Same transitions as before
- ▶ Allows to distinguish between death before / after relapse

Impact of relapse on survival probability

- ▶ What impact does relapse have on survival probability?
- ▶ Does this impact change with time?
- ▶ Do the effects of the baseline risk factors change after relapse?
- ▶ How probable is it to be alive after relapse has taken place?

NB: descriptive model, causal interpretation limited!

Event counts

Based on 724 patients without missing covariates:

State from	Nr entering	State to			
		2	3	4	No event
SCT	724	170	242	–	312
Rel	170	–	–	109	61
NRM	242	–	–	–	–
Death after rel	109	–	–	–	–

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Some terminology

- ▶ In the previous picture, a box is called a **state**
- ▶ A direct arrow from one state to another is a **transition**
 - ▶ States from which no transitions are possible are called **absorbing** states
 - ▶ States that are not absorbing are called **transient** states
- ▶ Special cases of multi-state models:
 - ▶ Survival analysis with one end-point (one initial, one absorbing state)
 - ▶ Competing risks (one initial, multiple absorbing states)
 - ▶ Illness-death model (simplest non-trivial multi-state model)
- ▶ Note: we assume that we observe the exact transition times, subject to right censoring (and/or left truncation)

Notation

- ▶ State space $\mathcal{S} = \{1, \dots, S\}$, finite
- ▶ Random process $\{X(t); t \geq 0\}$ taking values in $\mathcal{S}$
- ▶ History of the process until s : $\mathcal{H}_s = \{X(u); 0 \leq u \leq s\}$
- ▶ **Transition probability:**

$$P(X(t) = \ell \mid X(s) = k, \mathcal{H}_{s-})$$

- ▶ This is the probability that the process is in state ℓ at time t , given that it was in state k at time s and conditionally on the past trajectory until time s

Markov property; Chapman-Kolmogorov

- ▶ The multi-state process is a **Markov process** if it fulfills the **Markov property**:

$$P(X(t) = \ell \mid X(s) = k, \mathcal{H}_{s-}) = P(X(t) = \ell \mid X(s) = k)$$

Future behavior of the process depends on the history only through the present

- ▶ Gather these transition probabilities in an $S \times S$ matrix

$$\mathbf{P}(s, t) = \{P_{k\ell}(s, t)\}$$

- ▶ The transition probability matrices satisfy the **Chapman-Kolmogorov** equations:

$$\mathbf{P}(s, t) = \mathbf{P}(s, u)\mathbf{P}(u, t), \quad s < u < t$$

The Chapman-Kolmogorov equations

- ▶ Recall from probability theory

$$P(A) = \sum_j P(A \cap B_j), \text{ when } \cap_j B_j = \Omega, \quad P(A \cap B_j) = P(A | B_j)P(B_j)$$

- ▶ Then we have

$$\begin{aligned} P_{gh}(s, t) &= P(X(t) = h | X(s) = g) = \\ &= \sum_j P(X(t) = h, X(u) = j | X(s) = g) = \end{aligned}$$

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$$\begin{aligned} P_{gh}(s, t) &= P(X(t) = h | X(s) = g) = \\ &= \sum_j P(X(t) = h, X(u) = j | X(s) = g) = \\ &= \sum_j P(X(t) = h | X(u) = j, X(s) = g) P(X(u) = j | X(s) = g) = \\ &\stackrel{(\text{Markov})}{=} \sum_j P(X(t) = h | X(u) = j) P(X(u) = j | X(s) = g) = \\ &= \sum_j P_{gj}(s, u) P_{jh}(u, t) \end{aligned}$$

Transition hazards

- ▶ **Transition hazards** or **transition intensities**:

$$h_{k\ell}(t) = \lim_{\Delta t \downarrow 0} \frac{P(X(t + \Delta t) = \ell \mid X(t) = k)}{\Delta t}$$

- ▶ **Cumulative transition hazards**:

$$H_{k\ell}(t) = \int_0^t h_{k\ell}(s) ds$$

- ▶ Total rate out of state k is the sum of $h_{k\ell}(t)$ over all states ℓ that can be reached from k ($\ell \leftarrow k$)

$$h_{k\bullet}(t) = \sum_{\ell \leftarrow k} h_{k\ell}(t)$$

- ▶ All this is meaningfully defined for Markov multi-state processes
- ▶ Transition hazard $h_{k\ell}(t)$ depends only on overall time t (**clock forward**)

Semi-Markov processes

- Define T_k to be the entry time in state k ; $X(t)$ is a **semi-Markov process** if

$$P(X(t) = \ell \mid X(s) = k, \mathcal{H}_{s-}) = P(X(t) = \ell \mid X(s) = k, T_k)$$

- Corresponding transition hazards:

$$h_{k\ell}(t; T_k) = \lim_{\Delta t \downarrow 0} \frac{P(X(t + \Delta t) = \ell \mid X(t) = k, T_k)}{\Delta t}$$

- So the transition hazards $h_{k\ell}(t; T_k)$ depend on two time scales, the overall time t and the **sojourn time** $t - T_k$
- A further simplification can be obtained by saying that the transition intensities depend only on the time spent in the current states: $h_{k\ell}(t; T_k) = h_{k\ell}(t - T_k)$
- These processes are also called **Markov renewal** or **clock reset** processes because after entry in a new state the clock is reset

Back to competing risks

Side step

- ▶ We have used transition hazards in competing risks
- ▶ Recall the **cause-specific hazards**

$$h_k(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \leq T \leq t + \Delta t, D = k \mid T \geq t)}{\Delta t}$$

- ▶ $X(t)$ jointly describes (T, D)
- ▶ $X(t) = 0 \Leftrightarrow T \geq t$
- ▶ $X(t) = k \Leftrightarrow T \leq t, D = k$

$$P(X(t+\Delta t) = k \mid X(t) = 0) = P(t \leq T \leq t+\Delta t, D = k \mid T \geq t)$$



Likelihood in special multi-state models

Ordinary survival

- ▶ Data: (t_i, d_i) , $d_i = 0, 1$
- ▶ Likelihood contribution:

$$h(t_i)^{d_i} S(t_i), \quad S(t_i) = \exp(-H(t_i))$$

Competing risks

- ▶ Data: (t_i, d_i) , $d_i = 0, 1, \dots, K$
- ▶ Likelihood contribution (total hazard $h_{\bullet}(t) = \sum_{k=1}^K h_k(t)$):

$$\begin{aligned} & h_1(t_i)^{\mathbf{1}\{d_i=1\}} \dots h_K(t_i)^{\mathbf{1}\{d_i=K\}} S(t_i), \quad S(t_i) = \exp(-H_{\bullet}(t_i)) \\ &= \prod_{k=1}^K \left\{ h_k(t_i)^{\mathbf{1}\{d_i=k\}} \exp(-H_k(t_i)) \right\} \end{aligned}$$

Data for general multi-state models

Case of exactly observed transition times

- ▶ (Intermittently observed, interval censored, case is discussed later in this course)
- ▶ Transition times: $(t_{i0}, t_{i1}, \dots, t_{i,n_i})$
 - ▶ First time, t_{i0} , is start of observation period
 - ▶ Usually, $t_{i0} = 0$
 - ▶ Last time, t_{i,n_i} , can be end of observation period (in case the last state is absorbing)
 - ▶ Otherwise there is also a **censoring time** $c_i > t_{i,n_i}$
- ▶ States: $(x_{i0}, x_{i1}, \dots, x_{i,n_i})$
 - ▶ Very often, all subjects start in the same state x_{i0} , for instance in state 1

Likelihood

Likelihood contribution

$$\begin{aligned}
 & \prod_{\ell \leftarrow x_{i0}} h_{x_{i0}, \ell}(t_{i1})^{\mathbf{1}\{x_{i1}=\ell\}} \exp\left(-\left(H_{x_{i0}, \bullet}(t_{i1}) - H_{x_{i0}, \bullet}(t_{i0})\right)\right) \\
 & \cdot \prod_{\ell \leftarrow x_{i1}} h_{x_{i1}, \ell}(t_{i2})^{\mathbf{1}\{x_{i2}=\ell\}} \exp\left(-\left(H_{x_{i1}, \bullet}(t_{i2}) - H_{x_{i1}, \bullet}(t_{i1})\right)\right) \\
 & \quad \dots \\
 & \cdot \prod_{\ell \leftarrow x_{i, n_i-1}} h_{x_{i, n_i-1}, \ell}(t_{i, n_i})^{\mathbf{1}\{x_{i, n_i}=\ell\}} \\
 & \quad \cdot \exp\left(-\left(H_{x_{i, n_i-1}, \bullet}(t_{i, n_i}) - H_{x_{i, n_i-1}, \bullet}(t_{i, n_i-1})\right)\right) \\
 & \quad \cdot \exp\left(-\left(H_{x_{i, n_i-1}, \bullet}(c_i) - H_{x_{i, n_i-1}, \bullet}(t_{i, n_i})\right)\right)
 \end{aligned}$$

Likelihood

- ▶ The likelihood is a product of terms describing observed transitions
- ▶ Each of these terms is itself a product over all transitions that are possible from the starting state
 - ▶ Like in the case of competing risks!
 - ▶ Delayed entry (left truncation)
- ▶ The last term (in gray) is not there if the multi-state process ends in an absorbing state

Data representation

Wide format

- ▶ This is the typical "original" format
- ▶ One line per patient, with time and status for each of the intermediate and final events

Long format

- ▶ Most convenient representation of data is in long format
- ▶ Instead of one line per patient, multiple lines, one for each transition for which the patient is at risk
- ▶ With starting and stopping time
- ▶ Following the structure of the likelihood

The mstate package

- ▶ R package for multi-state models
- ▶ Functions for data preparation, estimation and prediction (non-parametric and semi-parametric) in general multi-state models
- ▶ Implements the standard methods for multi-state models
- ▶ Functions for competing risks also available

Function *msprep* prepares data in long format

Original data (wide format) ...

	id	srv_mo	srv_s	rel_mo	rel_s
1	1	133.870173	0	85.204132	1
2	2	39.967132	0	39.967132	0
3	3	8.085456	1	8.085456	0

Function *msprep* prepares data in long format

Original data (wide format) ...

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1	1	133.870173	0	85.204132	1
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3	3	8.085456	1	8.085456	0

... in long format:

	id	from	to	trans	Tstart	Tstop	time	status
1	1	1	2	1	0	85.204132	85.204132	1
2	1	1	3	2	0	85.204132	85.204132	0
3	1	2	3	3	85.204132	133.870173	48.666041	0
4	2	1	2	1	0	39.967132	39.967132	0
5	2	1	3	2	0	39.967132	39.967132	0
6	3	1	2	1	0	8.085456	8.085456	0
7	3	1	3	2	0	8.085456	8.085456	1

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Estimation of transition hazards

- ▶ Non-parametric estimate

$$\Delta \hat{H}_{k\ell}(t) = dN_{k\ell}(t) / Y_k(t) ,$$

with

- ▶ $dN_{k\ell}(t)$ = number of observed $k \rightarrow \ell$ transitions at time t
- ▶ $Y_k(t)$ = number in state k at time t (at risk for $k \rightarrow \ell$ transition)
- ▶ Straightforward extension of Nelson-Aalen estimator
- ▶ $N_{k\ell}(t)$ is a counting process, counting the number of $k \rightarrow \ell$ transitions $\leq t$

Estimation of transition hazards

Delayed entry

- ▶ For the transition Relapse → Death we have to account for delayed entry (left truncation)
 - ▶ Patients become at risk for Rel → Death only after they enter state 2 (Rel)

Variance estimates

- ▶ Analogous to univariate survival analysis
 - ▶ Aalen:

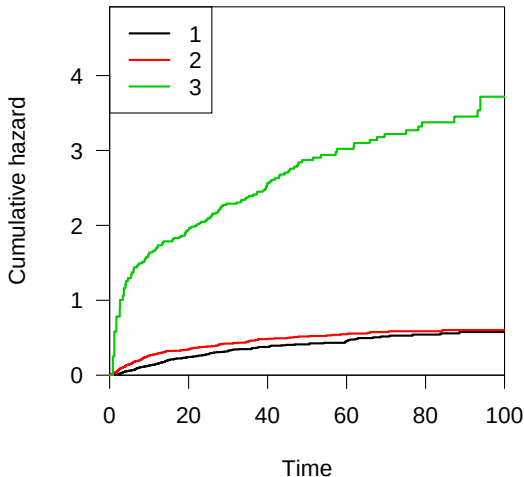
$$\widehat{\text{var}}(\Delta \hat{H}_{k\ell}(t)) = dN_{k\ell}(t) / Y_k^2(t), \text{ if } k \neq \ell$$

- ▶ Greenwood:

$$\widehat{\text{var}}(\Delta \hat{H}_{k\ell}(t)) = \frac{dN_{k\ell}(t)}{Y_k(t) \cdot (Y_k(t) - 1)}, \text{ if } k \neq \ell$$

Cumulative hazards

Non-parametric model



Transition probabilities

- ▶ Prediction time s (after transplant, surgery, diagnosis)
- ▶ Suppose we are in state k at time s
- ▶ What is the probability that we are in state ℓ at time t ?
- ▶ This is the **transition probability**

$$P_{k\ell}(s, t) = P(X(t) = \ell \mid X(s) = k)$$

- ▶ Only makes sense for a Markov process, otherwise $P(X(t) = \ell \mid X(s) = k, \{X(u); 0 \leq u < s\})$ may not equal $P(X(t) = \ell \mid X(s) = k)$
- ▶ Gather these transition probabilities in an $S \times S$ matrix

$$\mathbf{P}(s, t) = \{P_{k\ell}(s, t)\}$$

Kolmogorov equations

- ▶ Gather transition intensities $h_{k\ell}(t)$, $h_{kk}(t) = -\sum_{\ell \neq k} h_{k\ell}(t)$, into $S \times S$ matrix $\mathbf{h}(t) = \{h_{k\ell}(t)\}$
- ▶ Similarly for the cumulative transition hazards $H_{k\ell}(t)$ in $\mathbf{H}(t)$
- ▶ $\mathbf{P}(s, t)$ is the unique solution of the **Kolmogorov forward differential equations**

$$\frac{\partial}{\partial t} \mathbf{P}(s, t) = \mathbf{P}(s, t) \mathbf{h}(t)$$

with initial condition $\mathbf{P}(s, s) = \mathbf{I}$

- ▶ By a general result for product-integrals (Volterra's equation), this solution takes the form

$$\mathbf{P}(s, t) = \prod_{(s, t]} (\mathbf{I} + d\mathbf{H}(u))$$

- ▶ The $\prod$ is a product-integral

Aalen-Johansen

- ▶ As a result, a reasonable estimator of $\mathbf{P}(s, t)$ is given by

$$\hat{\mathbf{P}}(s, t) = \prod_{s < u \leq t} (\mathbf{I} + \Delta \hat{\mathbf{H}}(u))$$

- ▶ where $\hat{\mathbf{H}}(u) = \{\hat{H}_{k\ell}(u)\}$ is a matrix with all estimated transition hazards and on the diagonal

$$\hat{H}_{kk}(u) = -\sum_{\ell \neq k} \hat{H}_{k\ell}(u)$$

- ▶ This is called the Aalen-Johansen estimator (1978)
- ▶ Computational aspects and how to do this in `mstate` will be discussed on the last day

Special cases

- ▶ For special cases of multi-state models we may get explicit, and sometimes well-known, formulas for the elements of $\hat{\mathbf{P}}(s, t)$
- ▶ Ordinary survival analysis
 - ▶ Two states, 0=alive, 1=dead, $\hat{P}_{00}(0, t)$ is Kaplan-Meier

$$\hat{S}(t) = \prod_{u \leq t} \left(1 - \hat{P}(\text{failing at } u \mid \text{alive at } u-) \right)$$

- ▶ Competing risks
 - ▶ $K + 1$ states, 0=alive, $k > 1$ =dead of cause k , $\hat{P}_{0k}(0, t)$ is the cumulative incidence

Exercise

- ▶ Check that Kaplan-Meier and cumulative incidence are special cases of the Aalen-Johansen formula

The illness-death model

Illness-death model, 0=alive, 1=ill, 2=dead

$$\hat{P}_{11}(s, t) = \prod_{s < u \leq t} (1 - d\hat{H}_{12}(u))$$

$$\hat{P}_{12}(s, t) = \sum_{s < u \leq t} \hat{P}_{11}(s, u-) d\hat{H}_{12}(u)$$

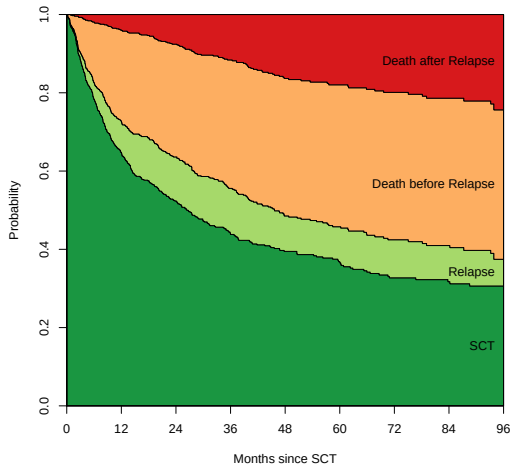
$$\hat{P}_{00}(s, t) = \prod_{s < u \leq t} (1 - d\hat{H}_{01}(u) - d\hat{H}_{02}(u))$$

$$\hat{P}_{01}(s, t) = \sum_{s < u \leq t} \hat{P}_{00}(s, u-) d\hat{H}_{01}(u) \hat{P}_{11}(u+, t)$$

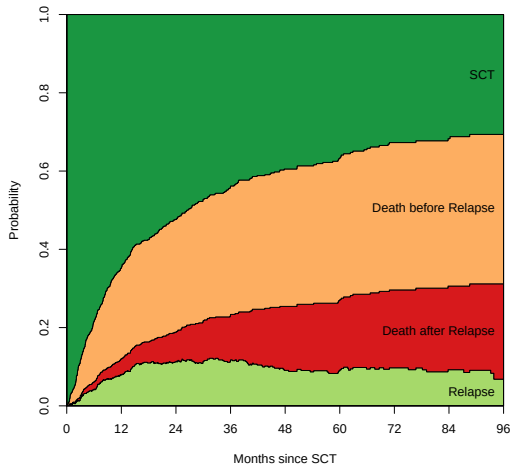
$$\hat{P}_{02}(s, t) = \sum_{s < u \leq t} \hat{P}_{00}(s, u-) d\hat{H}_{01}(u) \hat{P}_{12}(u+, t)$$

$$+ \sum_{s < u \leq t} \hat{P}_{00}(s, u-) d\hat{H}_{02}(u)$$

Transition probabilities I

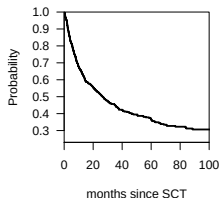


Transition probabilities II

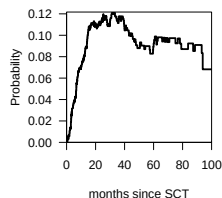


Transition probabilities III

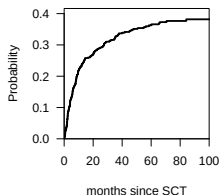
SCT



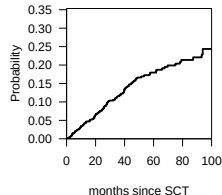
Relapse



NRM



Death after Relapse



Summary

- ▶ Concepts: transition intensities, transition probabilities, state occupation probabilities
- ▶ Central relation between transition intensities and probabilities
- ▶ Non-parametric estimation
 - ▶ Extension of Nelson-Aalen estimate of cumulative transition intensities
 - ▶ Aalen-Johansen estimator of transition probabilities
- ▶ Next lecture: how to incorporate covariates

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